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1 Accessibility of healthcare facilities and malaria prevalence in Kenya

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Course: Fundamentals of Spatial Data Engineering

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2 Contents

3 Assignment

3.1 0) PACKAGES

```
# Install once if needed:
# install.packages(c("sf", "raster", "sp", "spdep", "gstat", "gridExtra", "cowplot"))

library(sf)
library(raster)
library(sp)
library(spdep)
library(gstat)
library(gridExtra)
library(cowplot)
```

3.1.1 0.1) LOAD DATA

```
setwd("C:/Users/darey/OneDrive/Documents/MGEO/MGEO/FromLinearRegressionToGeostatisticalRegression/Data/S")

# Data per tutorial
rain.DF <- read.table("sic_obs.dat", sep = ",", header = TRUE)
Boundary.sf <- st_read("Swiss_Bound.shp")

## Reading layer `Swiss_Bound' from data source
## `C:/Users/darey/OneDrive/Documents/MGEO/MGEO/FromLinearRegressionToGeostatisticalRegression/Data/S'
## using driver `ESRI Shapefile'
## Simple feature collection with 1 feature and 2 fields
## Geometry type: POLYGON
## Dimension: XY
## Bounding box: xmin: -169990.4 ymin: -111695.5 xmax: 177125.7 ymax: 108158.3
## CRS: NA

Boundary.sp <- as(Boundary.sf, "Spatial")
DEM <- raster("DEM.grd")
NDVI <- raster("NDVI.grd")

rain.SpDF <- rain.DF
coordinates(rain.SpDF) <- c("x_coor", "y_coor")

# Independent vars from rasters at station locations
x1 <- extract(DEM, rain.SpDF, df = TRUE)[,2] # Elevation
x2 <- extract(NDVI, rain.SpDF, df = TRUE)[,2] # NDVI
y <- rain.SpDF@data$rain_mm # Rainfall (response)

# Design matrix (intercept, DEM, NDVI)
X <- cbind(1, x1, x2)
```

```
# Store back into attributes for convenience
rain.SpDF@data$y <- y
rain.SpDF@data$x1 <- x1
rain.SpDF@data$x2 <- x2
```

4 Task 1 — Compare simple linear models: $y \sim x1$ vs $y \sim x2$

Fit two separate linear regression models with either elevation or NDVI. Compare the two models with the appropriate metric and draw conclusions on which is the best.

```
#Get simple linear model for elevation.
```

```
LM_x1 <- lm(y ~ x1)
summary(LM_x1)
```

```
##
## Call:
## lm(formula = y ~ x1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -241.26  -80.33  -12.82   62.91  326.04
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  899.8262   102.9763   8.738 6.59e-14 ***
## x1           -0.8289    0.1181  -7.019 2.93e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 95.67 on 98 degrees of freedom
## Multiple R-squared:  0.3345, Adjusted R-squared:  0.3278
## F-statistic: 49.27 on 1 and 98 DF,  p-value: 2.934e-10
```

```
#Get simple linear model for NDVI.
```

```
LM_x2 <- lm(y ~ x2)
summary(LM_x2)
```

```
##
## Call:
## lm(formula = y ~ x2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -158.407  -51.286   -0.666   47.090  185.084
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    28.77     12.99   2.214  0.0291 *
## x2             513.05     37.39  13.720 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 68.62 on 98 degrees of freedom
## Multiple R-squared:  0.6576, Adjusted R-squared:  0.6541
## F-statistic: 188.3 on 1 and 98 DF,  p-value: < 2.2e-16

# Get Metrics
rmse <- function(m) sqrt(mean(residuals(m)^2))

# Compute metrics for both simple linear regressions
data.frame(
  Model    = c("y ~ x1 (DEM)", "y ~ x2 (NDVI)"),
  SSE      = c(sum(residuals(LM_x1)^2), sum(residuals(LM_x2)^2)),
  MSE      = c(mean(residuals(LM_x1)^2), mean(residuals(LM_x2)^2)),
  RMSE     = c(rmse(LM_x1), rmse(LM_x2)),
  R2       = c(summary(LM_x1)$r.squared, summary(LM_x2)$r.squared),
  Adj_R2   = c(summary(LM_x1)$adj.r.squared, summary(LM_x2)$adj.r.squared)
)

##           Model      SSE      MSE      RMSE      R2      Adj_R2
## 1 y ~ x1 (DEM) 896920.4 8969.204 94.70588 0.3345462 0.3277559
## 2 y ~ x2 (NDVI) 461439.5 4614.395 67.92934 0.6576433 0.6541499
```

4.1 Discussion: Comparing Simple Linear Models

Two simple linear regression models were fitted to evaluate how well each independent variable explains the variation in rainfall.

4.1.1 Model 1: $y \sim x_1$ (DEM)

The regression equation is:

$$\text{textRainfall} = 899.83 - 0.8289 \text{time} \text{textElevation}$$

The coefficient for elevation is negative, indicating that rainfall decreases by approximately **0.83 mm for every 1-meter increase in elevation**. This negative relationship is statistically significant ($p = 2.93\text{e-}10$), showing that elevation does influence rainfall, but the relationship is inverse.

- $R^2 = 0.3345$, meaning elevation explains about **33%** of the rainfall variability.
- Adjusted $R^2 = 0.3278$
- RMSE = **94.71 mm**, which represents the average prediction error.

4.1.2 Model 2: $y \sim x_2$ (NDVI)

The regression equation is:

$$\text{textRainfall} = 28.77 + 513.05 \text{time} \text{textNDVI}$$

The slope is positive and highly significant ($p < 2\text{e-}16$), meaning rainfall increases with vegetation density. Areas with higher NDVI values (greener vegetation) tend to have higher rainfall.

- $R^2 = 0.6576$, indicating that NDVI alone explains about **66%** of the rainfall variability.
- Adjusted $R^2 = 0.6541$
- RMSE = **67.93 mm**, much smaller than the elevation model's RMSE.

4.1.3 Model comparison

Metric	y ~ x1 (DEM)	y ~ x2 (NDVI)	Interpretation
SSE	896,920.4	461,439.5	NDVI model has smaller total squared error
MSE	8,969.2	4,614.4	NDVI model has lower average error variance
RMSE	94.71	67.93	NDVI predictions are more accurate
R²	0.3345	0.6576	NDVI explains nearly twice as much variance
Adjusted R²	0.3278	0.6541	NDVI still stronger after adjustment

4.1.4 Interpretation

The NDVI model performs substantially better in all metrics. Its R² value is about twice that of the elevation model, and the RMSE is roughly **27 mm lower**, meaning the predictions are more accurate. Although elevation is statistically significant, it explains a much smaller portion of rainfall variability.

Between the two simple linear regressions, **NDVI (y ~ x2) is the better predictor of rainfall**. It shows a strong positive and statistically significant relationship, higher explanatory power (R² = 0.66), and lower prediction error (RMSE = 68 mm).

5 Task 2 — Multiple OLS model (y ~ x1 + x2) and compare to Task 1

Fit a third model with both independent variables (elevation and NDVI) and make a comparison with the first two models. Which of the models is best (using the appropriate metric)?

```
# Fit model and inspect with both predictors
LM_both <- lm(y ~ x1 + x2)
summary(LM_both)
```

```
##
## Call:
## lm(formula = y ~ x1 + x2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -145.577  -42.041    3.736   39.440  187.258
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  291.19497   92.33266   3.154  0.00215 **
## x1          -0.27946    0.09742  -2.869  0.00506 **
## x2          445.95109   43.00347  10.370 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 66.22 on 97 degrees of freedom
## Multiple R-squared:  0.6844, Adjusted R-squared:  0.6779
## F-statistic: 105.2 on 2 and 97 DF,  p-value: < 2.2e-16
```

```
# Metrics (same style as Task 1)
rmse <- function(m) sqrt(mean(residuals(m)^2))

# Single-row metrics for the multiple model
data.frame(
  Model    = c("y ~ x1 (DEM)", "y ~ x2 (NDVI)", "y ~ x1 + x2"),
  SSE      = c(sum(residuals(LM_x1)^2),
               sum(residuals(LM_x2)^2),
               sum(residuals(LM_both)^2)),
  MSE      = c(mean(residuals(LM_x1)^2),
               mean(residuals(LM_x2)^2),
               mean(residuals(LM_both)^2)),
  RMSE     = c(rmse(LM_x1), rmse(LM_x2), rmse(LM_both)),
  R2       = c(summary(LM_x1)$r.squared,
               summary(LM_x2)$r.squared,
               summary(LM_both)$r.squared),
  Adj_R2   = c(summary(LM_x1)$adj.r.squared,
               summary(LM_x2)$adj.r.squared,
               summary(LM_both)$adj.r.squared)
)
```

```
##           Model      SSE      MSE      RMSE      R2      Adj_R2
## 1 y ~ x1 (DEM) 896920.4 8969.204 94.70588 0.3345462 0.3277559
## 2 y ~ x2 (NDVI) 461439.5 4614.395 67.92934 0.6576433 0.6541499
## 3 y ~ x1 + x2 425352.3 4253.523 65.21903 0.6844176 0.6779108
```

5.1 Discussion: Multiple Linear Regression ($y \sim x1 + x2$)

A multiple linear regression model was fitted using both elevation ($x1$) and NDVI ($x2$) as predictors of rainfall. The model equation is:

$$\text{textRainfall} = 291.19 - 0.2795\text{time}\text{textElevation} + 445.95\text{time}\text{textNDVI}$$

Both predictors are statistically significant ($p < 0.01$). The negative coefficient for elevation indicates that rainfall decreases slightly with higher altitude, while the positive coefficient for NDVI shows that rainfall increases with vegetation density. The intercept of approximately 291.19 mm represents the expected rainfall when both elevation and NDVI are near zero.

5.1.1 Model performance

- **$R^2 = 0.6844$** , meaning the combined model explains about **68% of the variability in rainfall**, a clear improvement compared to the single-variable models.
- **Adjusted $R^2 = 0.6779$** , confirming the model remains strong even after adjusting for the additional variable.
- **RMSE = 65.22 mm**, which is smaller than both simple regression models (94.71 mm for DEM and 67.93 mm for NDVI).
- **F-statistic = 105.2**, with a very low p-value ($< 2.2e-16$), indicating the model as a whole is highly significant.

Model	SSE	MSE	RMSE	R ²	Adjusted R ²
y ~ x1 (DEM)	896,920.4	8,969.2	94.71	0.3345	0.3278
y ~ x2 (NDVI)	461,439.5	4,614.4	67.93	0.6576	0.6541
y ~ x1 + x2	425,352.3	4,253.5	65.22	0.6844	0.6779

5.1.2 Interpretation

Including both elevation and NDVI improves the model's predictive power compared to using either variable alone. While NDVI remains the stronger predictor, adding elevation provides additional explanatory information.

6 Task 3 — Compare RMSE of OLS (Task 2) vs Geostatistical Regression (GLS)

Using the RMSE of the linear regression model in Task 2 and that of a geostatistical regression model, which of the two models has a better predictive power?

```
# 1) Start from OLS residuals and fit a variogram
e_ols <- residuals(LM_both)
vgm.Emp.e <- variogram(e_ols ~ 1, locations = rain.SpDF)
vgmSph.e <- fit.variogram(vgm.Emp.e, model = vgm(psill = 8000, model = "Sph", range = 60000, nugget = 500))

# 2) Build initial covariance Σ from residual variogram and do first GLS
DMat <- spDists(rain.SpDF)
Sigma_ols1 <- variogramLine(vgmSph.e, dist_vector = DMat, covariance = TRUE)
Beta_gls1 <- solve(t(X) %*% solve(Sigma_ols1) %*% X) %*% (t(X) %*% solve(Sigma_ols1) %*% y)
e_gls1 <- as.vector(y - (X %*% Beta_gls1))

# 3) Iterate: re-fit variogram on GLS residuals, rebuild Σ, update coefficients
K_iter <- 100
B.gls.ite <- matrix(NA_real_, nrow = K_iter, ncol = ncol(X))
for (i in 1:K_iter) {
  v_emp <- variogram(e_gls1 ~ 1, locations = rain.SpDF, covariogram = FALSE)
  v_fit <- fit.variogram(v_emp, model = vgm(8000, "Sph", 60000, nugget = 500))
  Sigma <- variogramLine(v_fit, dist_vector = DMat, covariance = TRUE)
  B.gls.ite[i, ] <- solve(t(X) %*% solve(Sigma) %*% X) %*% (t(X) %*% solve(Sigma) %*% y)
  e_gls1 <- as.vector(y - (X %*% B.gls.ite[i, ]))
}

# 4) Final GLS coefficients and residuals
B.gls <- B.gls.ite[K_iter, ]
e.gls <- as.vector(y - (X %*% B.gls))

# 5) RMSE (using residual degrees of freedom)
n <- length(y); k <- 2
SSE_ols <- t(e_ols) %*% e_ols
MSE_ols <- SSE_ols / (n - (k + 1))
RMSE_ols <- sqrt(MSE_ols)

SSE_gls <- t(e.gls) %*% e.gls
```

```

MSE_gls <- SSE_gls / (n - (k + 1))
RMSE_gls <- sqrt(MSE_gls)

data.frame(
  Model = c("OLS (y ~ x1 + x2)", "GLS (y ~ x1 + x2)"),
  RMSE = c(RMSE_ols, RMSE_gls)
)

```

```

##           Model      RMSE
## 1 OLS (y ~ x1 + x2) 66.2199
## 2 GLS (y ~ x1 + x2) 67.4369

```

6.1 Discussion: OLS vs GLS

Interpret the RMSE comparison. When might GLS help, and why might it not here?

7 Task 4 — CV comparison: Ordinary Kriging (OK) vs Regression Kriging (RK)

Using leave-one-out cross-validation, compare the predictive power of ordinary kriging and regression kriging. Use the elevation and NDVI as your independent variables.

4.1) Fit a variogram model for rainfall (for OK & RK)

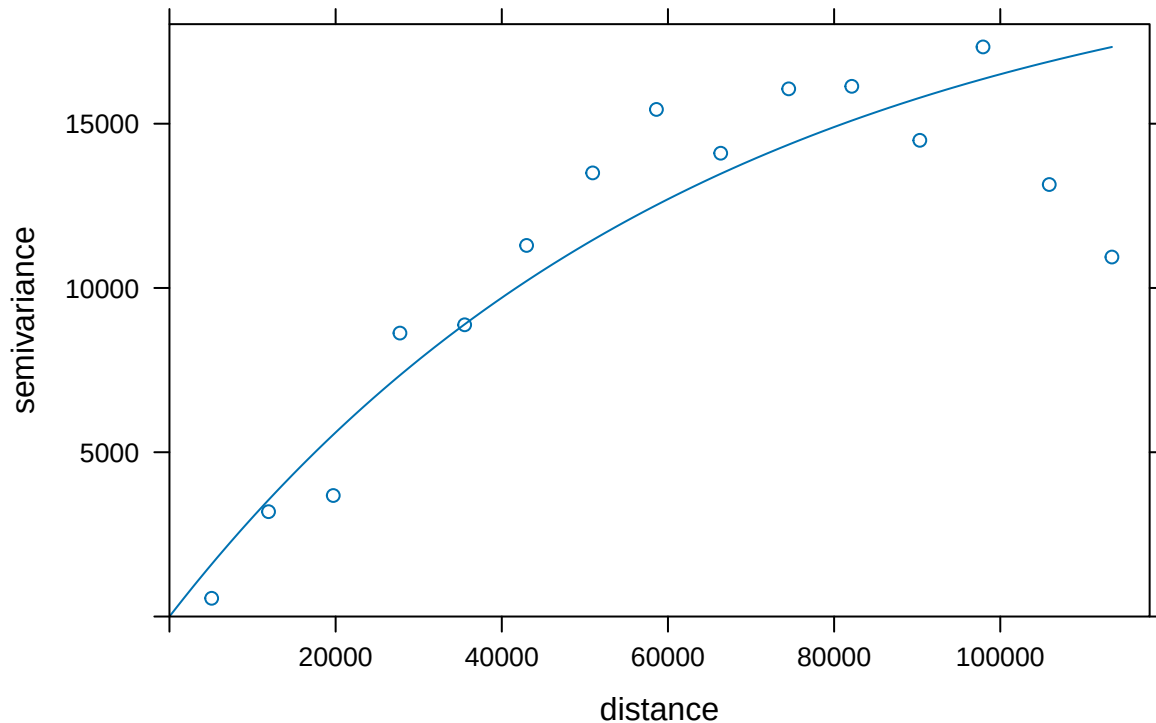
```

#### 4) VARIOGRAM FOR RAINFALL (used by OK & RK) ####
vgm.Emp.y <- variogram(y ~ 1, locations = rain.SpDF)
vgmExp.y <- fit.variogram(vgm.Emp.y, model = vgm(psill = 15000, model = "Exp",
                                                range = 60000, nugget = 500))

# Inspect:
plot(vgm.Emp.y, vgmExp.y, main="Empirical vs Exponential variogram (Rainfall)")

```


Empirical vs Exponential variogram (Rainfall)



4.2) Leave-one-out CV: OK vs RK

4.2) CROSS-VALIDATION

Ordinary Kriging CV

```
OK.cv <- krige.cv(y ~ 1, locations = rain.SpDF, model = vgmExp.y)
```

```
## |
```

Regression Kriging CV (mean uses covariates)

```
RK.cv <- krige.cv(y ~ x1 + x2, locations = rain.SpDF, model = vgmExp.y)
```

```
## |
```

Metrics function

```
cv_metrics <- function(cv_obj, y_obs = rain.SpDF$y) {
  res <- cv_obj$residual
  ME <- mean(res) # mean error (bias)
  MSE <- mean(res^2)
  RMSE <- sqrt(MSE)
  R2 <- 1 - var(res) / var(y_obs) # proportion of explained variance
  data.frame(ME=ME, MSE=MSE, RMSE=RMSE, R2=R2)
}
```

```
OK_metrics <- cv_metrics(OK.cv)
```

```
RK_metrics <- cv_metrics(RK.cv)
```

```

rbind(
  cbind(Model="Ordinary Kriging", OK_metrics),
  cbind(Model="Regression Kriging", RK_metrics)
)

```

```

##           Model      ME      MSE      RMSE      R2
## 1 Ordinary Kriging -2.091845 4689.439 68.47948 0.6524003
## 2 Regression Kriging -2.185775 4799.854 69.28098 0.6442380

```

7.1 Discussion: OK vs RK

Summarize ME/RMSE/R² differences and when RK is expected to outperform OK.

8 References

Add your .bib entries in references.bib. Any cited works will render here automatically.